

Milestone 1

1. The first program is top. Top displays all the processes running on the machine, also showing how much of the cpu and memory each process is using. The top program is sorted by the CPU usage. However, we can sort by memory usage, PID, and other parameters. For that we need to press f while top is running, and choose what needs to be sorted. Knowing what processes use the most of the cpu and memory is important in performance. With help of top, we also could detect saturation, even before it happening.
2. Vmstat measures the memory usage, CPU activity, disk I/O, paging and interrupts. The output is not sorted as in top, so it is presented as a snapshot. Vmstat is helpful for performance because it helps us detect excessive swapping, which would suggest that memory is full. Vmstat also can show if there's high CPU wait times or heavy I/O activity that may slow down the system
3. Iostat measures disk I/O performance and CPU utilization. The output displays the storage devices and shows different stats for each driver: read/s, write/s, Kb read, Kb wrote. The output of iostat is not sorted. The iostat program helps us detect the issues related to the disk.
4. Nstat measures the network performance. It displays packets sent, packets received, errors and dropped packets. The nstat displays the counter for each network event, rather than a sorted table. With nstat we can detect the problems that are caused by the network.

5. Perf measures low-level CPU utilization such as processor cycles, executed instructions, cache misses and other hardware performance events. There are multiple commands that can be used with top, each displaying different stats. The output is not sorted. The numbers we get from perf can help us detect CPU bottlenecks, excessive cache misses and other issues that could slow down the system.